Comparing Mutual Information in Finite Samples

Can MI-specific degrees of freedom improve comparisons between populations?

Daniel Zhao (*Master of Computer Science*)

Supervisor: Associate Professor Joseph T. Lizier

The University of Sydney



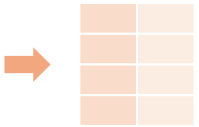
Motivation: comparing MI across populations

Mutual information measures how much observing one variable reduces uncertainty about another, on average.
Different sample estimates can arise from sampling noise.

Null hypothesis $H_0 : I(P) = I(Q)$

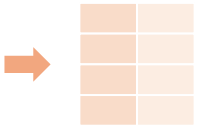
Two populations

Population P
sample size n_P



MI estimate
$\hat{I}(P)$

Population Q
sample size n_Q



MI estimate
$\hat{I}(Q)$

Variables: two categorical variables, X and Y.

Statistical task

Summarise a sample from each population in a count table

Estimate MI from each table

Assess whether sampling noise can explain the difference

Does MI differ between the two populations?

Table icons are schematic. Thesis Sections 1.1 and 2.1; Bossomaier et al. (2016)

Literature review: foundations and the MI-specific step

Existing results provide MI estimates, a Wald test using a standard normal distribution, and moment matching.
This project derives the MI variance sensitivity needed for its degrees of freedom.

	MI estimation and uncertainty	*Wald comparison*	*Welch–Satterthwaite moment matching*
Idea	Estimate MI and first-order sampling variance	Divide the estimated difference by its standard error	Match variance moments to choose degrees of freedom
Key work	Moddemeijer (1989); Paninski (2003)	Wald (1943); Mora and Ruiz-Castillo (2009)	Satterthwaite (1946); Welch (1947); Hutcheson (1970); Pan and Wall (2002)
Required step	Track variation in the variance estimate itself	Assess the fixed cutoff from the standard normal distribution in finite samples	Calculate the MI-specific variance sensitivity

Contribution: Derive that sensitivity and evaluate the resulting degrees of freedom.

Thesis Sections 2.2–2.5 and 3.2–3.3; full references in B13–B14

Method: one statistic, two reference distributions

Both methods use **the same estimated MI difference and standard error**.
Expanded Welch uses estimated degrees of freedom for a Student t distribution.

Shared statistic

$$T = \frac{\widehat{\Delta}_{\mathrm{BC}}}{\widehat{\mathrm{SE}}}, \qquad \widehat{\mathrm{SE}}^2 = \frac{\widehat{V}(P)}{n_P} + \frac{\widehat{V}(Q)}{n_Q}$$

Δ_{BC}: estimated $I(P) - I(Q)$, after leading bias correction.

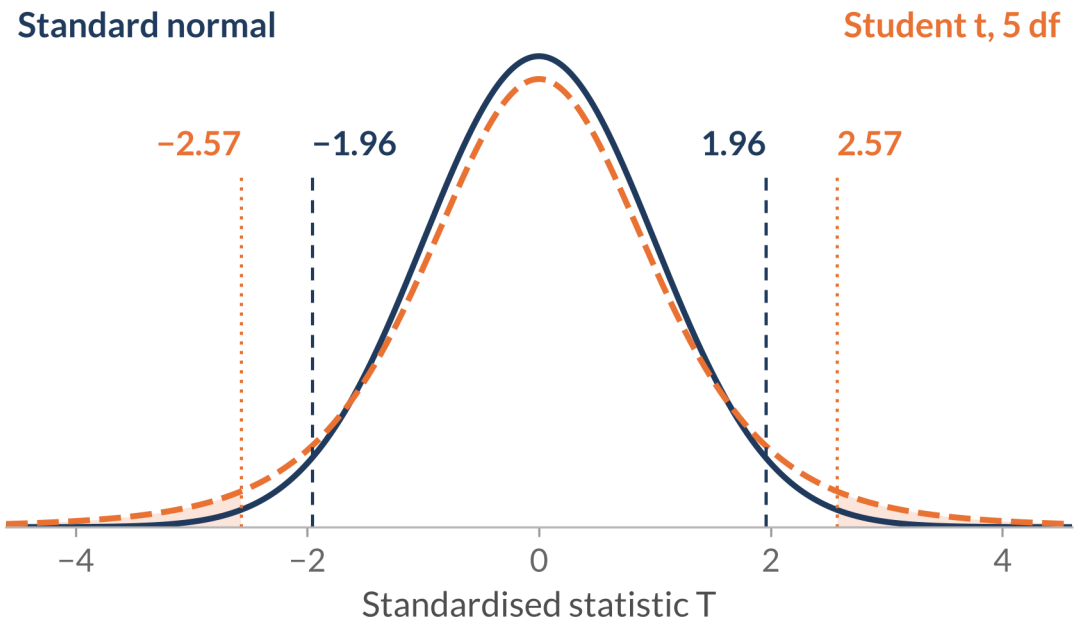
$V(P)$: population variance of pointwise MI;
$\widehat{V}(P)$ estimates it from one sampled table.

Wald test

Standard normal distribution; two-sided 5% cutoff about **±1.96**.

Expanded Welch test

Student t distribution with estimated degrees of freedom; **a larger cutoff**.



Student t distribution with 5 degrees of freedom, for illustration; cutoff about ±2.57.

Approach and contribution

The estimated pointwise-MI variance changes between samples.
We approximate that variation and **use it to choose the degrees of freedom**.

Derivation chain

Cell probabilities change
Move a small amount of probability towards one cell.

→

Margins and pointwise MI change
Track their joint effect.

→

Variance sensitivity, derived here
$g_P(x, y)$ describes the change in $V(P)$.

→

Degrees of freedom
Use the spread of those sensitivities.

Result and approximations

$$\mathrm{Var}\{\widehat{V}(P)\} \approx \frac{\tau^2(P)}{n_P} \quad \Longrightarrow \quad \hat{\nu}_P = \frac{2n_P\widehat{V}(P)^2}{\hat{\tau}^2(P)}$$

$\tau^2(P)$ is the variance of $g_P(X, Y)$. Repeat for Q and combine using Welch–Satterthwaite.

Two approximations evaluated

1. First-order sampling expansion
2. Matching a scaled chi-squared distribution

Thesis Sections 4.1–4.3; exact derivative in B1, moment matching in B2

Evaluation design

Known populations separate approximation accuracy, false positives, and power.
Both tests receive **the same sampled table pairs**.

Research questions

1. **RQ1: Approximation accuracy**
   Sampling moments of $\hat{V}$ and component df, including near independence
2. **RQ2: False positives**
   Equal MI at 0.005, 0.02, and 0.30 nats
3. **RQ3: Power**
   $I(P)$ = 0.02; MI-gap magnitudes 0.005, 0.01, and 0.02 nats

Shared simulation grid

- Four sizes: 2×2, 4×4, 8×8, 16×16
- Uniform, same-skew, and different-skew margins
- Ten sample sizes: 10 to 50,000 observations per population
- 100,000 independent table pairs per setting

Model: copying construction: choose margins and solve for target MI.

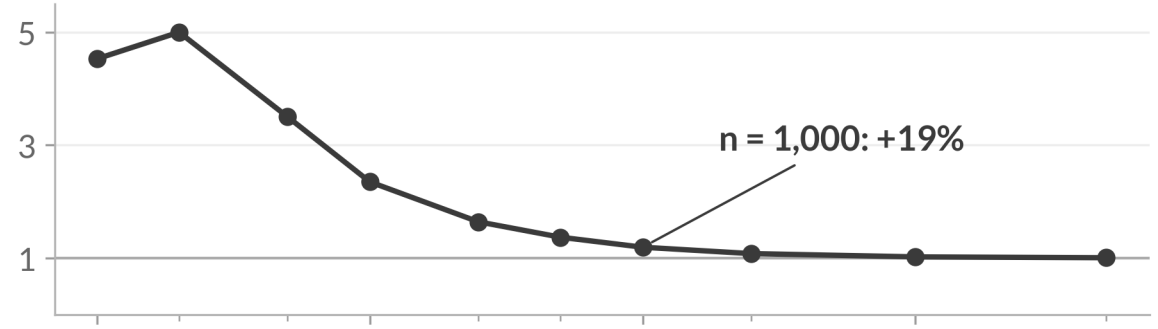
Reporting: usual 0.05 p-value thresholds. Unsuccessful calculations count as non-rejections; calculation success is reported separately.

Morris et al. (2019). Thesis Chapter 5, Tables 5.1–5.3; exact vectors, sample sizes, and requirements in B3

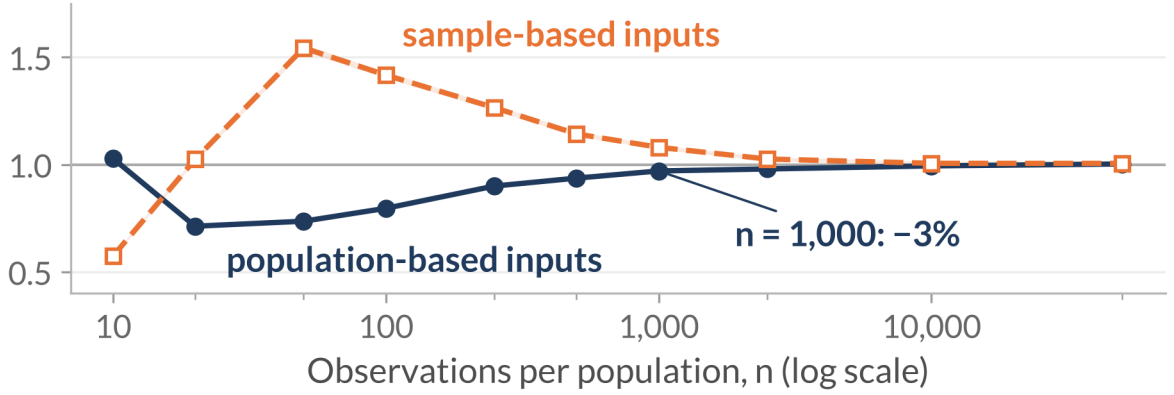
RQ1 results: approximation accuracy

The mean of $\hat{V}$ can **remain inflated** even when its population first-order sampling-variance prediction is accurate.

Mean of $\hat{V}$ / population V



Predicted / observed sampling variance of $\hat{V}$



Different-skew P · 4×4 · population MI 0.02 nats

Mean: rare cells can inflate it.

Newly observed rare cells can have large pointwise-MI deviations, whose squares raise $\hat{V}$.

Sampling variance: linearisation can miss large changes.

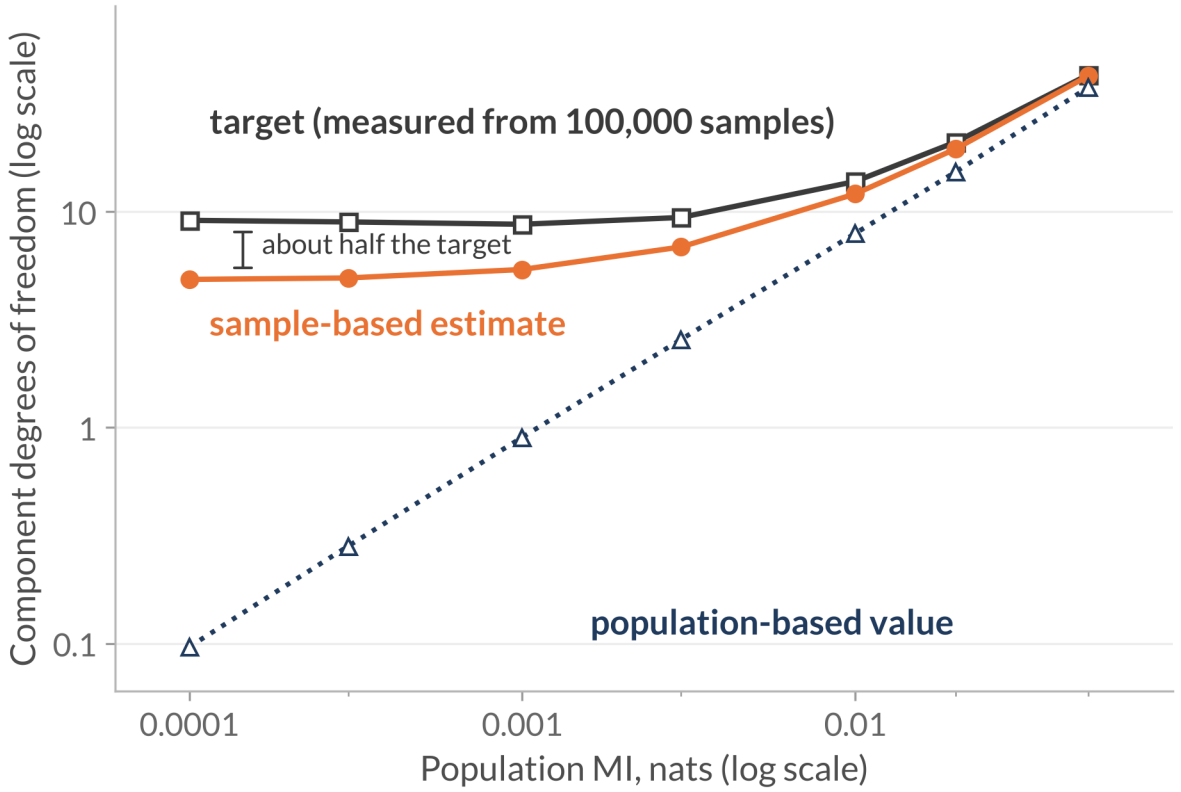
Rare cells appear in some samples and not others; sample-based sensitivities can also overpredict the resulting variation.

At n = 1,000: mean **19% high**;
variance prediction **3% low** with population-based inputs and **8% high** with sample-based inputs.
Both moments enter the degrees of freedom.

Thesis Sections 6.1.1–6.1.2, Figures 6.1–6.2. Bands: approximate 95% Monte Carlo intervals. Other sizes and Q: B4

RQ1 results: near independence

Near independence, first-order theory gives almost no degrees of freedom, while **sampling noise keeps the sample-based estimate at about half the target**.



Target: component degrees of freedom from the actual mean and variance of V across 100,000 samples.

Different-skew P · 4×4 · n = 1,000 fixed

Population-based: falls to about n × MI

Inserts known population inputs into the first-order formula. Near independence it is about n × population MI, which heads to zero.

Sample-based: noise keeps it near 5

Quadratic sampling errors create apparent association, which remains in each sample's calculation.

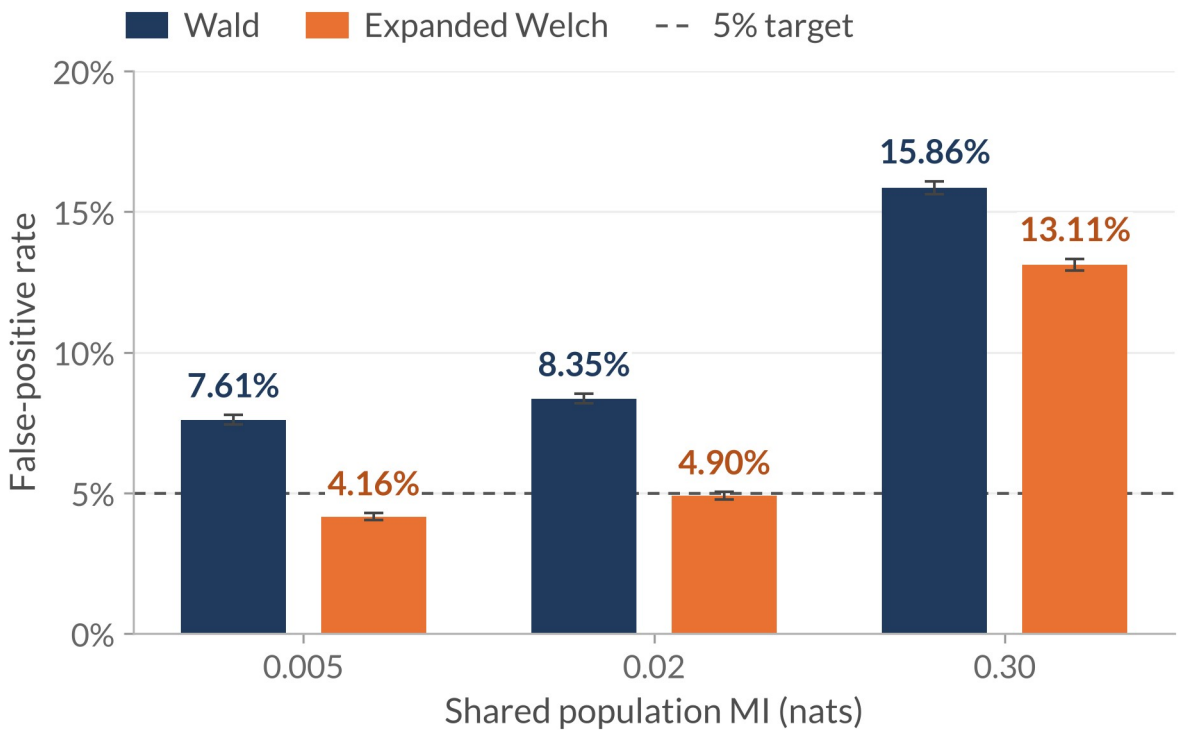
At MI 0.0001: target **≈9.1**; sample-based estimate **≈4.9**; population-based value **≈0.1**

Thesis Section 6.1.3, Figure 6.3; Appendix A.2. Lines: target $\nu_{P,\mathrm{mom}}$, mean of $\hat{\nu}_P$, population-based $\nu_{P,\mathrm{FO}}$. Bands: approximate 95% Monte Carlo intervals. Local calculation: B5

RQ2 results: false positives across shared MI values

A larger cutoff **brings some excessive false-positive rates closer to 5%**, but accuracy at one association level does not persist across shared MI values.

Uniform 4×4 · n = 20 per population · equal MI in P and Q



Shared MI 0.02: closer to 5%

Expanded Welch lowers the rate from 8.35% to 4.90%.

Shared MI 0.30: still above 5%

Expanded Welch still rejects in 13.11% of repetitions.

Interpretation: The larger cutoff reduces rejection, while the shared statistic can still have the wrong sampling distribution.

Nominal target: 5%. Both tests' calculations succeed in every repetition in these examples.

Slices of the balanced null grid. Whiskers: pointwise 95% Monte Carlo intervals. Thesis Section 6.2.1, Figure 6.4; full grids in B6–B8

RQ2 results: why bias changes rejection

Similar errors in estimated uncertainty accompany **opposite false-positive behaviour** when bias in the MI difference changes.

16×16 · n = 1,000 per population · equal MI 0.02 nats

Margins	Variance ratio	Bias / SD	Wald	Expanded Welch
Same skew	1.93	≈0	0.60%	0.53%
Different skew	1.94	**2.06**	26.01%	24.84%

Variance ratio: average squared estimated standard error / observed variance of the estimated MI difference.

Same skew

The oversized denominator suppresses the statistic; expected MI biases cancel.

Different skew

The estimated difference is shifted away from zero by unequal residual biases.

> **Conclusion:** Changing the cutoff leaves the biased estimated difference unchanged.
> Both tests' calculations succeed in every repetition here.

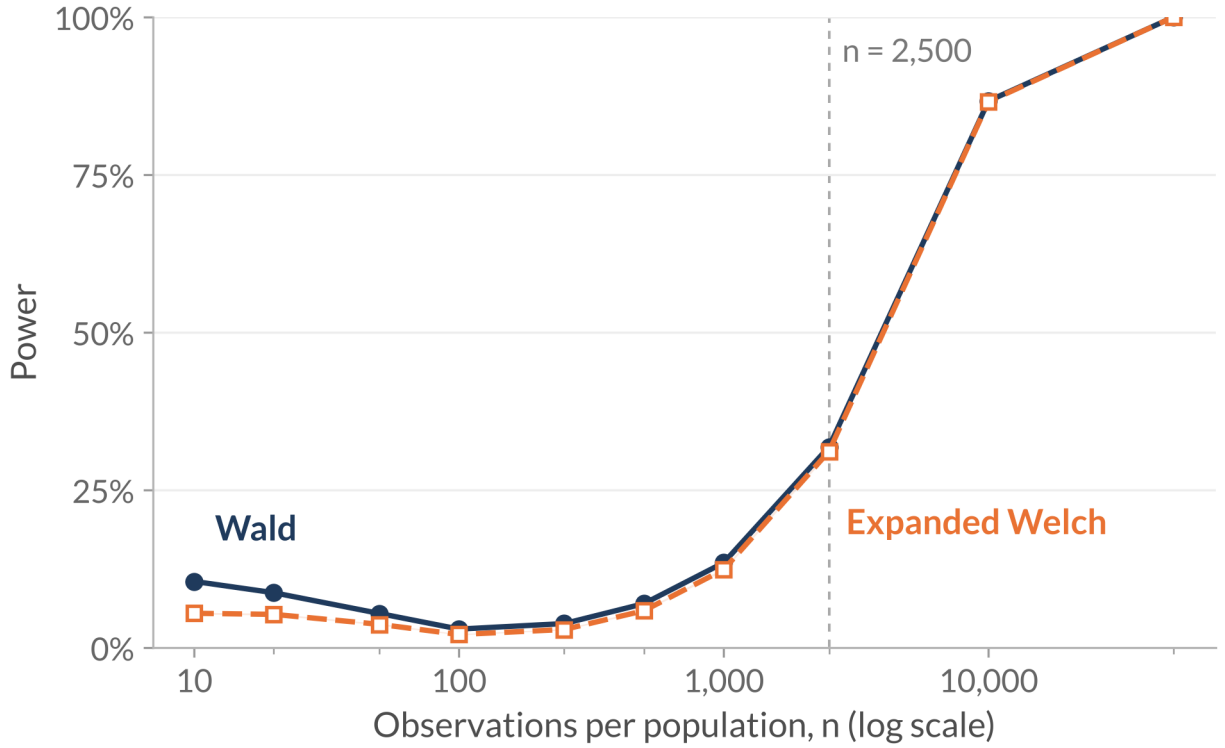
Thesis Section 6.2.4, Table 6.1. Full different-skew grid: B8; ordinary Welch comparison: B9

RQ3 results: power across MI differences

With sufficient observations, **larger MI differences become easier to detect**.
Expanded Welch's larger cutoff lowers or preserves rejection.

Uniform 4×4 · population MI in P = 0.02 nats

MI-gap magnitude 0.01 nats: power against sample size



Snapshot at n = 2,500

MI-gap magnitude (nats)	Wald	Expanded Welch
0.005	11.60%	11.14%
0.01	31.85%	31.14%
0.02	79.24%	78.74%

Power: probability of rejecting equal MI when population MI differs. Rates use the usual 0.05 p-value threshold.

Rates divide rejections by all 100,000 repetitions; unsuccessful calculations count as non-rejections.

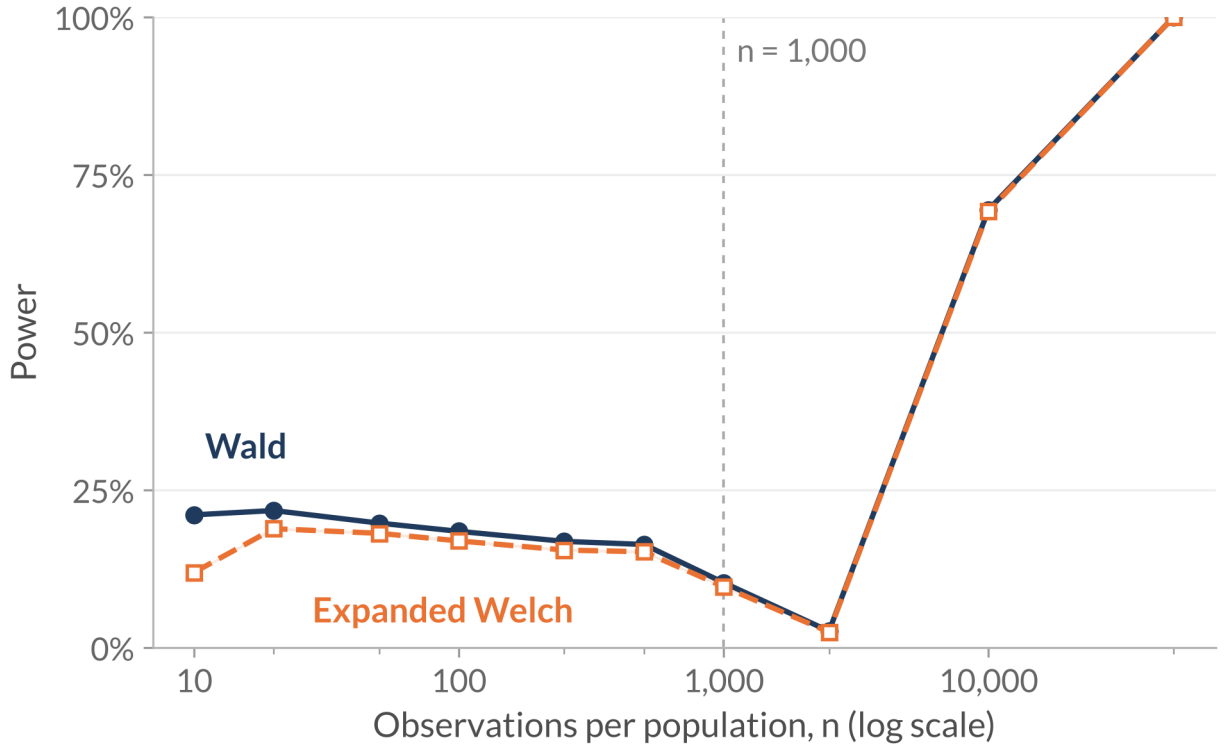
Bands: pointwise 95% Monte Carlo intervals. Thesis Section 6.3.1, Figure 6.7; full grids in B10–B12

RQ3 results: bias can reverse the estimated direction

In sparse tables, unequal residual biases can outweigh the true MI difference and **reverse the estimated direction**.

Different-skew 16×16 · MI 0.02 versus 0.03 nats

MI-gap magnitude 0.01 nats: power against sample size



At n = 1,000

True difference, P minus Q	**−0.0100 nats**
Mean estimated difference	**+0.0207 nats**

Expanded Welch rejects in **9.66%** of repetitions.

Among those rejections, over 99.9% have the opposite estimated direction.

Mechanism: As positive bias diminishes, the sampling centre passes near zero, so rejection initially falls before recovering.

At n = 1,000: 9,659 of 9,662 Expanded Welch rejections are positive; every calculation succeeds. Thesis Figure 6.9; full grid in B12

Conclusions and next steps

MI-specific degrees of freedom improve some false-positive rates, while remaining failures identify **estimation accuracy as a priority**.

Contribution and findings

1. Derived variance sensitivity and MI-specific degrees of freedom.
2. Brought some excessive false-positive rates closer to 5%; deepened conservatism elsewhere.
3. Lower power under the same 5% testing rules; residual bias can reverse the estimated direction.

> **Scope:** The evaluation covers one population construction and equal sample sizes, with the Wald test as the main benchmark.

Next steps

1. Reduce bias in the estimated MI difference and improve its standard error and degrees of freedom, including near independence.
2. Test other association constructions and unequal sample sizes.

Thank you.

Appendix

Variance sensitivity

Perturbation towards one cell: $P_\varepsilon = (1-\varepsilon)P + \varepsilon\delta_{(x,y)}$

Changing probability weights

$$g_P(x,y) = \{\ell_P(x,y) - I(P)\}^2 - V(P)$$

Changing pointwise-MI values and margins

$$+ 2[\ell_P(x,y) - \mathbb{E}_P\{\ell_P(X,Y) \mid X = x\}$$

$$- \mathbb{E}_P\{\ell_P(X,Y) \mid Y = y\} + I(P)]$$

Pointwise MI: $\ell_P(i,j) = \log\{p_{ij}/(p_{i+}p_{+j})\}$

Sensitivity variance: $\tau^2(P) = \mathrm{Var}_P\{g_P(X,Y)\}$

Exact for positive population tables; its use for sampling errors is first order. Thesis Chapter 4

Moment matching and combined degrees of freedom

Component degrees of freedom

$$\nu_{P,\,\mathrm{mom}} = \frac{2[\mathbb{E}\{\widehat{V}(P)\}]^2}{\mathrm{Var}\{\widehat{V}(P)\}}, \qquad \widehat{\nu}_P = \frac{2n_P\widehat{V}(P)^2}{\widehat{\tau}^2(P)}$$

Combined degrees of freedom

$$\widehat{\nu} = \frac{\{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q\}^2}{\{\widehat{V}(P)/n_P\}^2/\widehat{\nu}_P + \{\widehat{V}(Q)/n_Q\}^2/\widehat{\nu}_Q}$$

Moment matching does not establish the standard normal numerator, or its independence from the variance estimates, that an exact Student t distribution requires.

Satterthwaite (1946). Thesis Sections 4.1 and 4.3

Population construction, grid, and calculation requirements

$$p_{ij}(\lambda) = (1-\lambda)a_i a_j + \lambda a_i \mathbf{1}\{i=j\}, \qquad 0 \le \lambda < 1$$

Parameter grid

- k = 2, 4, 8, 16; uniform, same-skew, or different-skew margins
- Same skew: both use $m_k(0.8)$. Different skew: P uses $m_k(0.7)$, Q uses $m_k(0.8)$, where $m_k(\rho) = (\rho, (1-\rho)/(k-1), \ldots)$
- Null: shared MI 0.005, 0.02, 0.30. Power: P MI 0.02, Q MI 0.025, 0.03, 0.04

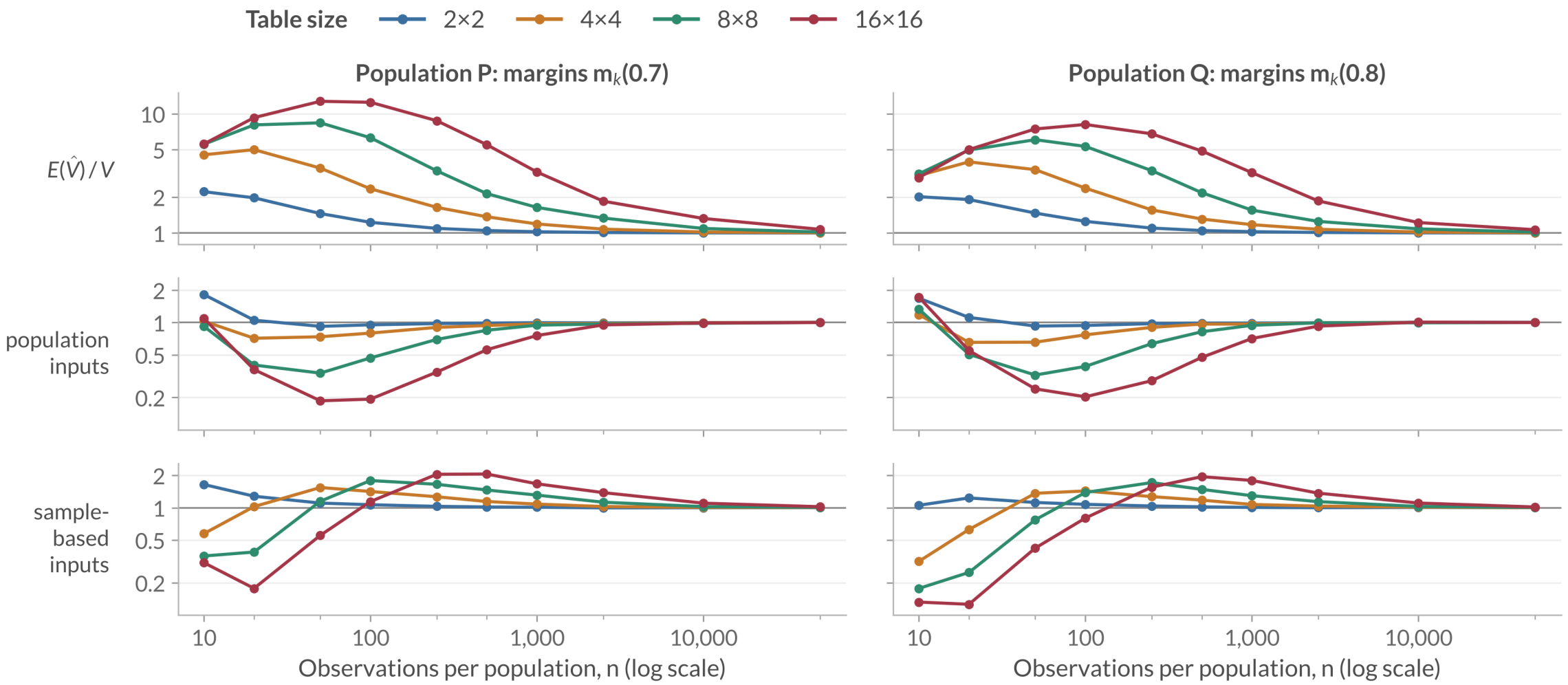
Calculation requirements and rates

- Each null or power study has 360 settings, 100,000 independent pairs per setting
- A Wald calculation needs a finite difference and a positive finite standard error. Expanded Welch also needs positive finite degrees of freedom.
- Rate = rejections from successful calculations / all repetitions. Calculation success is reported separately.

n = 10, 20, 50, 100, 250, 500, 1,000, 2,500, 10,000, 50,000 per group

Approximation ratios across table sizes

Different-skew margins, shared MI 0.02 nats; a ratio of one indicates agreement



Rows: mean of $\hat{V}/V$; population-input and sample-based variance predictions / observed variance. Thesis Figure 6.1

Near-independence noise calculation

Setting: 4×4, n = 1,000, MI 0.0001; $d = 9$ and $\kappa = 2nI(P) \approx 0.2$

$$G = 2n\hat{I}(P) \;\dot{\sim}\; \chi^2_d(\kappa), \qquad \hat{\nu}_P \approx G/2$$

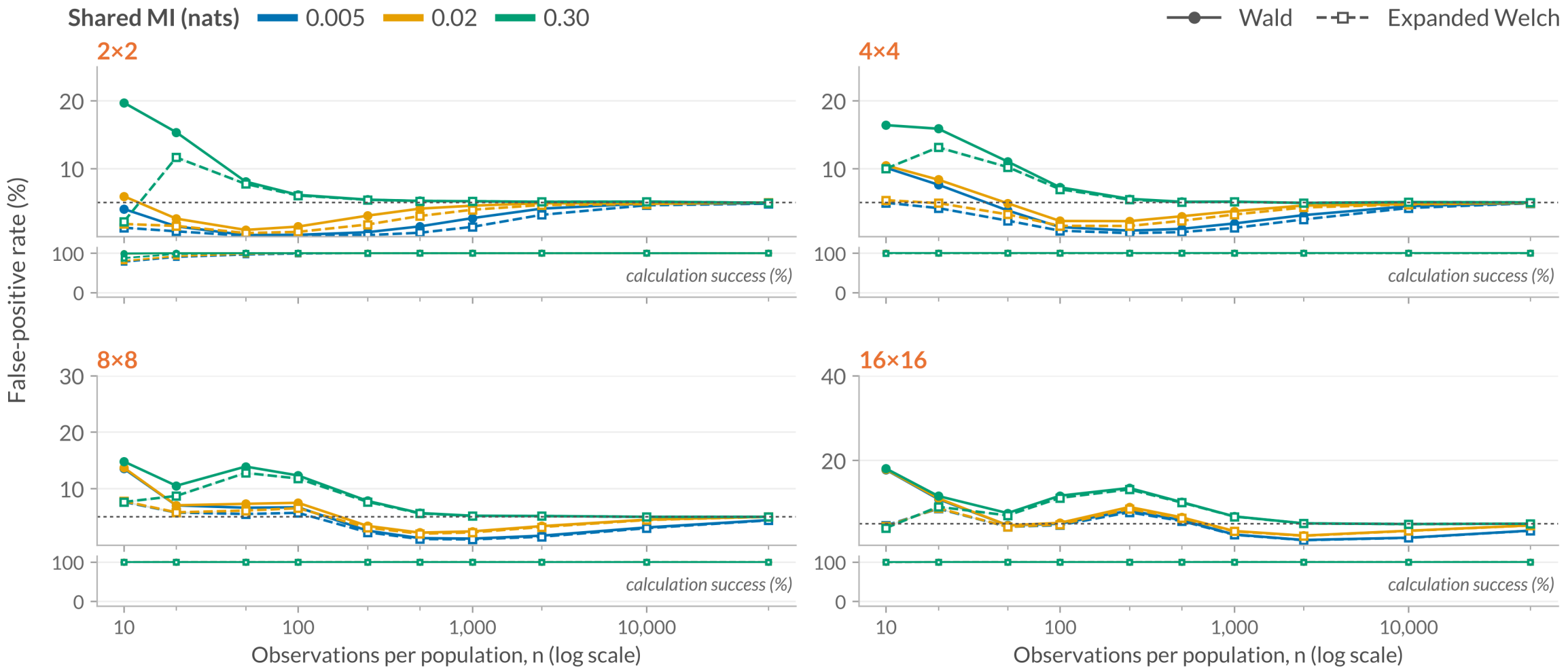
In $G, \hat{I}(P)$ is the uncorrected plug-in MI, not a bias-corrected estimate.

Quantity	Local prediction	Observed (P)
Mean estimated component df	4.60	4.86
Sampling-moment target	9.00	9.12
Mean sample-based variance prediction / observed sampling variance	1.96	1.97

Local approximations under fixed dimensions and positive limiting cell probabilities. Thesis Appendix A.2, Section 6.1.3

Full null results: uniform margins

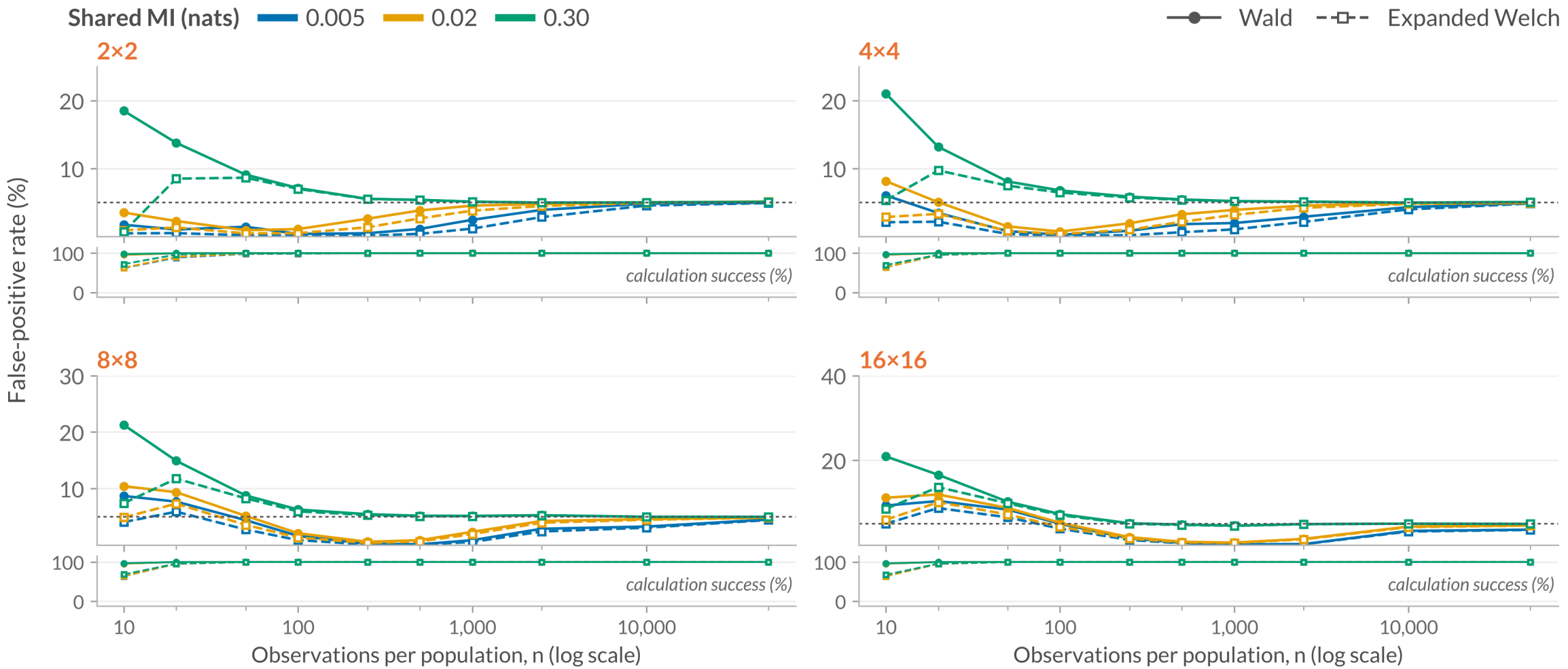
Equal MI in P and Q; uniform margins in both populations



Rates divide rejections by all 100,000 repetitions; unsuccessful calculations count as non-rejections. Requirements in B3. Thesis Figure 6.4

Full null results: same-skew margins

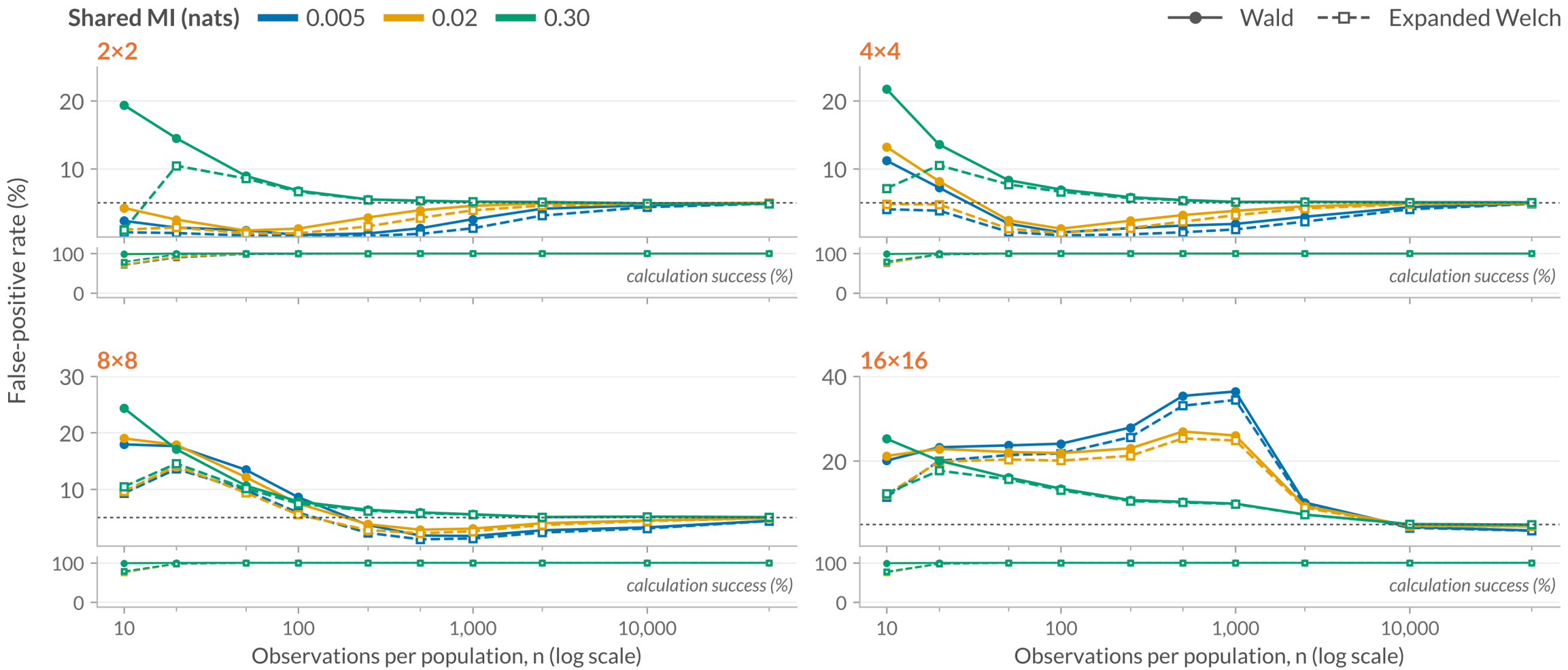
Equal MI; same-skew margins $m_k(0.8)$ in P and Q



Rates divide rejections by all 100,000 repetitions; unsuccessful calculations count as non-rejections. Requirements in B3. Thesis Figure 6.5

Full null results: different-skew margins

Equal MI; $m_k(0.7)$ in P and $m_k(0.8)$ in Q



Rates divide rejections by all 100,000 repetitions; unsuccessful calculations count as non-rejections. Requirements in B3. Thesis Figure 6.6

Ordinary Welch degrees of freedom

Same statistic and sampled pairs; ordinary Welch gives each component $n - 1$ degrees of freedom; shared MI 0.02 nats

Setting	Wald	Ordinary Welch	Expanded Welch
Uniform 4×4, n = 20	8.35%	7.35%	4.90%
Uniform 4×4, n = 50	4.87%	4.57%	3.24%
Different skew 16×16, n = 1,000	26.01%	25.95%	24.84%

False-positive rates; nominal target 5%.

n = 20

Ordinary Welch reduces the excess less than Expanded Welch.

n = 50

Wald is already close to 5%; Expanded Welch is the most conservative.

Different skew 16×16

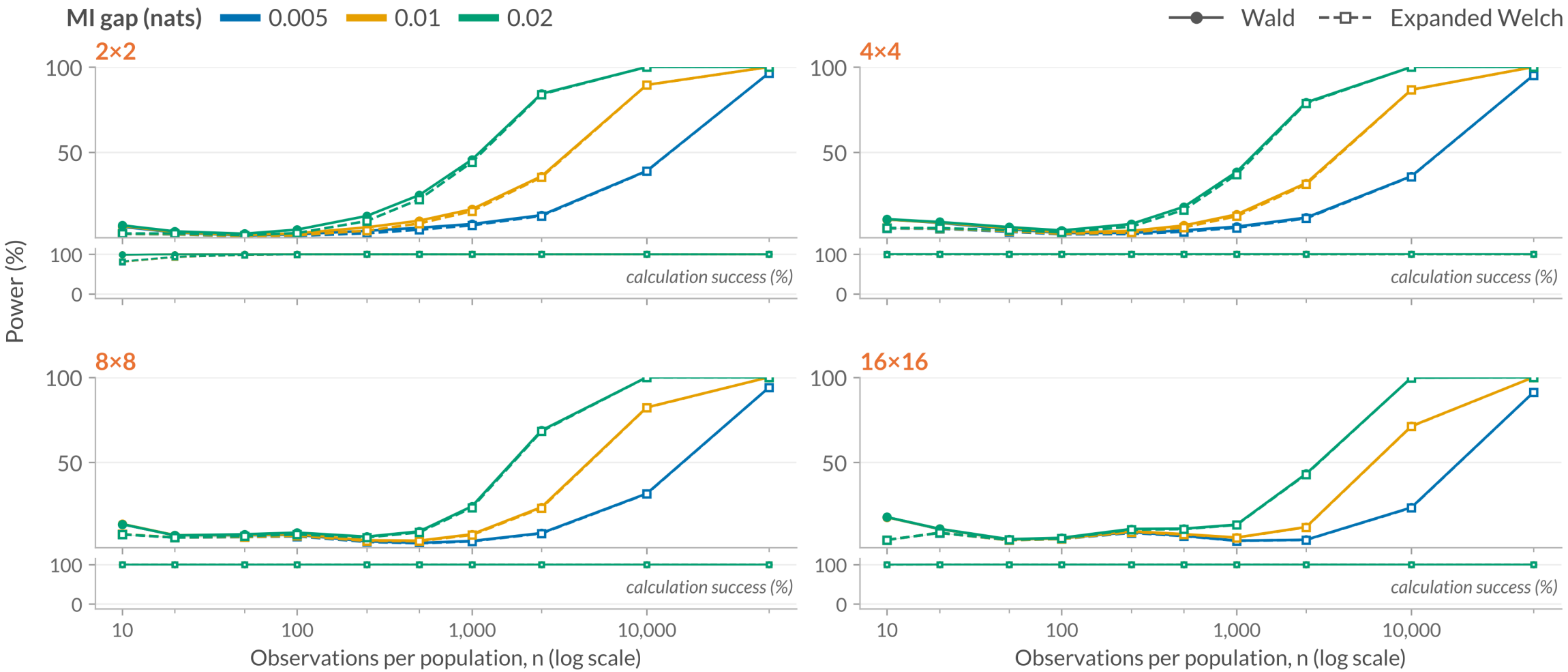
Neither choice removes the excess caused by bias.

Conclusion: MI-specific degrees of freedom change the results, but a larger reduction in rejection does not consistently improve calibration.

Same 100,000 sampled pairs per setting as the null study; every calculation succeeds. Thesis Section 6.2.5, Table 6.2

Full power results: uniform margins

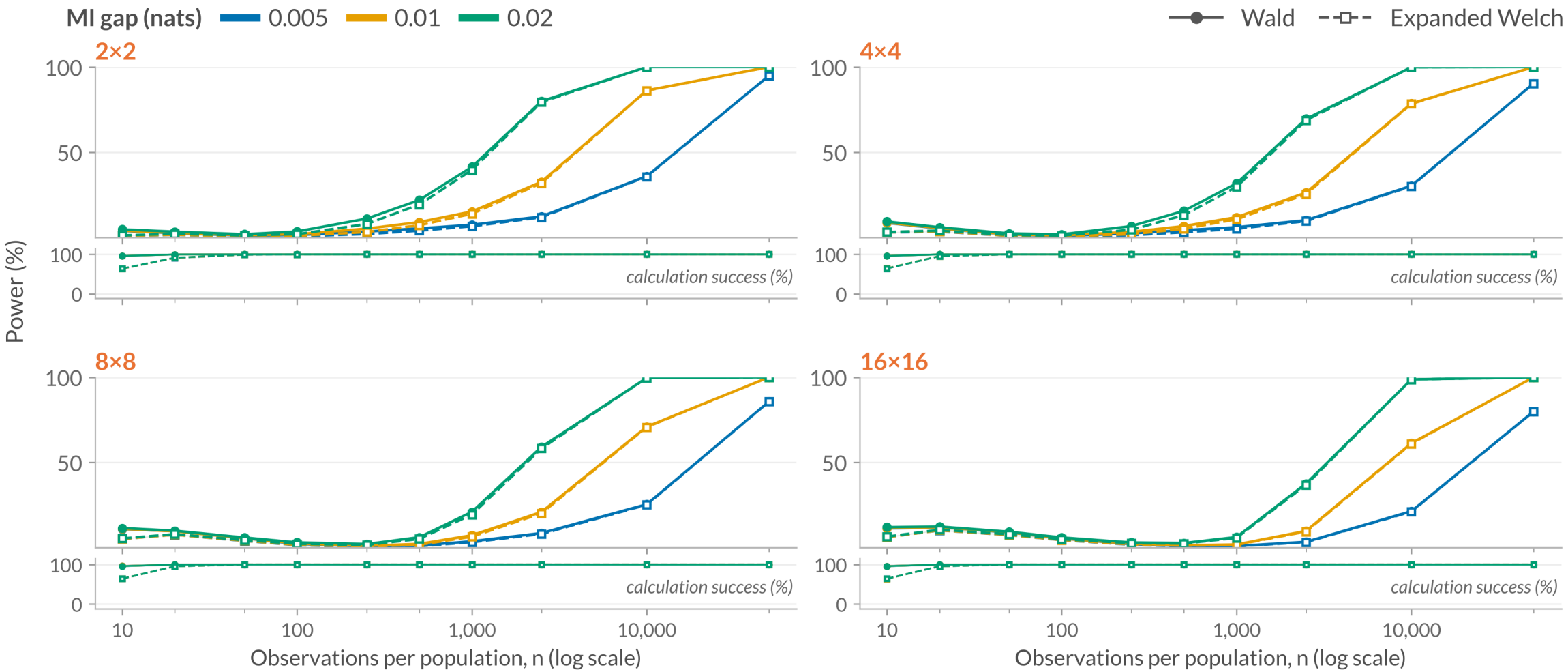
Uniform margins; MI in P = 0.02 nats



Rates divide rejections by all 100,000 repetitions; unsuccessful calculations count as non-rejections. Requirements in B3. Thesis Figure 6.7

Full power results: same-skew margins

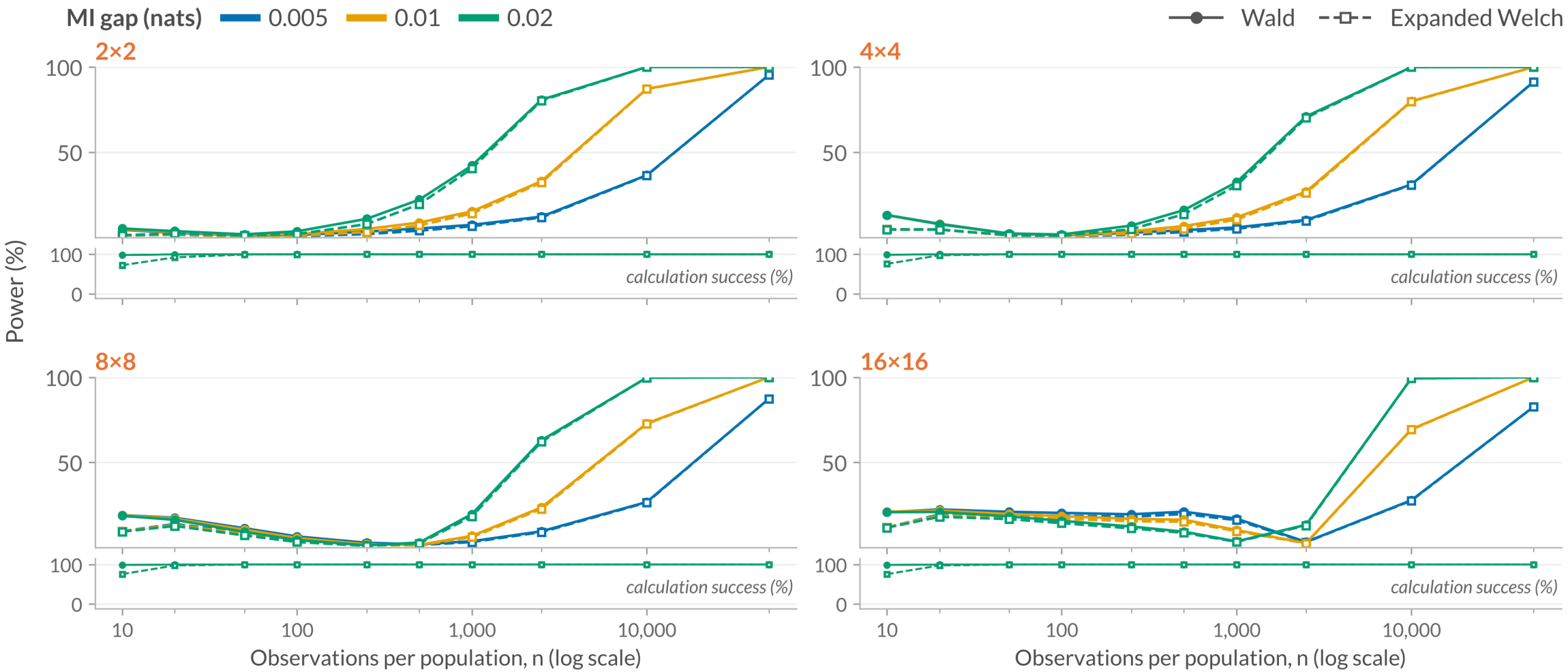
Same-skew margins $m_k(0.8)$ in P and Q; MI in P = 0.02 nats



Rates divide rejections by all 100,000 repetitions; unsuccessful calculations count as non-rejections. Requirements in B3. Thesis Figure 6.8

Full power results: different-skew margins

$m_k(0.7)$ in P, $m_k(0.8)$ in Q; MI in P = 0.02 nats



Rates divide rejections by all 100,000 repetitions; unsuccessful calculations count as non-rejections. Requirements in B3. Thesis Figure 6.9

References: MI and testing foundations

Bossomaier, T., Barnett, L., Harré, M. and Lizier, J. T. (2016). *An Introduction to Transfer Entropy: Information Flow in Complex Systems*. Springer, Cham. doi:10.1007/978-3-319-43222-9

Moddemeijer, R. (1989). On estimation of entropy and mutual information of continuous distributions. *Signal Processing*, 16(3), 233–248. doi:10.1016/0165-1684(89)90132-1

Paninski, L. (2003). Estimation of entropy and mutual information. *Neural Computation*, 15(6), 1191–1253. doi:10.1162/089976603321780272

Mora, R. and Ruiz-Castillo, J. (2009). The statistical properties of the mutual information index of multigroup segregation. Working Paper, Economic Series (48), 09-84, Universidad Carlos III de Madrid, Departamento de Economía.

Satterthwaite, F. E. (1946). An approximate distribution of estimates of variance components. *Biometrics Bulletin*, 2(6), 110–114. doi:10.2307/3002019

Welch, B. L. (1947). The generalization of “Student’s” problem when several different population variances are involved. *Biometrika*, 34(1–2), 28–35. doi:10.1093/biomet/34.1-2.28

References: related methods and evaluation

Hutcheson, K. (1970). A test for comparing diversities based on the Shannon formula. *Journal of Theoretical Biology*, 29(1), 151–154. doi:10.1016/0022-5193(70)90124-4

Morris, T. P., White, I. R. and Crowther, M. J. (2019). Using simulation studies to evaluate statistical methods. *Statistics in Medicine*, 38(11), 2074–2102. doi:10.1002/sim.8086

Pan, W. and Wall, M. M. (2002). Small-sample adjustments in using the sandwich variance estimator in generalized estimating equations. *Statistics in Medicine*, 21(10), 1429–1441. doi:10.1002/sim.1142

Wald, A. (1943). Tests of statistical hypotheses concerning several parameters when the number of observations is large. *Transactions of the American Mathematical Society*, 54(3), 426–482. doi:10.1090/S0002-9947-1943-0012401-3